

Climate Risk + Resilience Impact on New Jersey Bonds

Quantifying Climate Effects on Bond Premia

Introductions

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Project Overview

1

Define Scope

2

Collect Data

3

Regressions

4

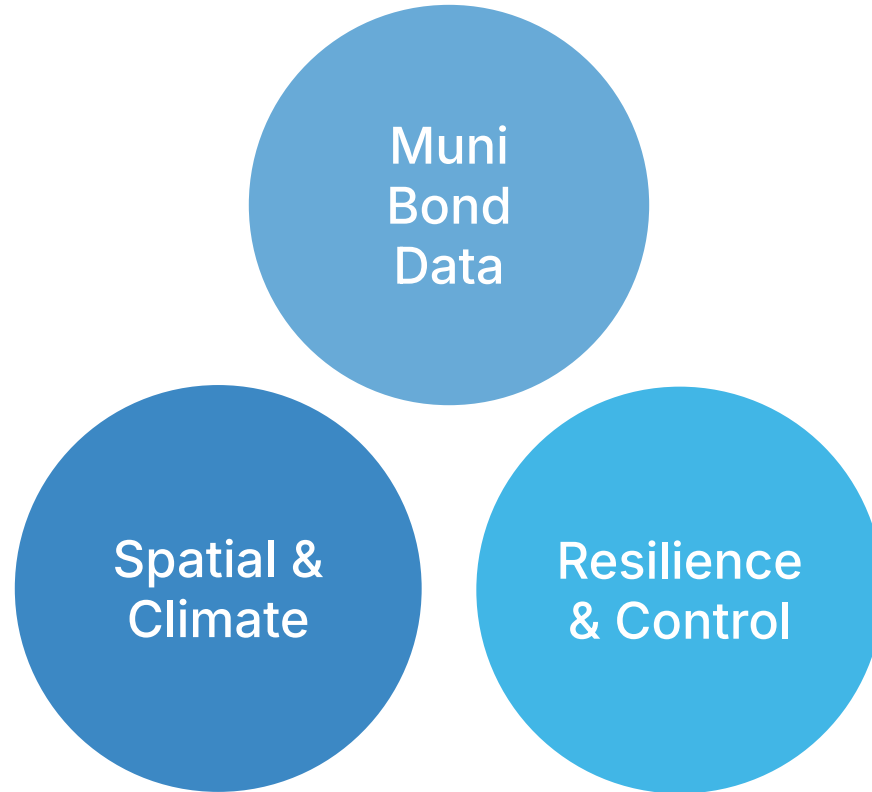
Conclusion

Project Scope

- Observation Period: 2015-2025
- Tracking all NJ Municipalities
 - GO Bond Premia
 - Flood Risk
 - State-Determined Resilience
 - Controlling Variables
- Extract Coefficients
 - Effect of Perceived Risk?
 - Importance of Resilience?

Data

Data Collection



Municipal Bonds Data

- Wharton Research Data Services
 - All US Bonds '15-'25
- Filter to NJ GO Bonds
 - Exclude Revenue/Special Tax Bonds
- Match CUSIP to Muni
 - Track trades over time
- Compute premium to AAA
 - FRED DGS[N]

Spatial Flood Data

- NJ FloodMapper
 - 2, 5, 7 ft. flood maps
- NJ Geographic Information Network
 - All NJ Tax Parcels
- Compute Intersections
 - Intersection $\rightarrow$ % Flooded
 - % Flooded $\rightarrow$ % Value Retained
 - Sum over municipalities
 - Interpolate for 4ft

Resilience & Controls

- Resilience Proxies
 - FEMA Community Rating System
 - MS4 and MLUL reports
 - NJ CHAMP Priority Scores
 - Flag for DID/Matched Pairs
- Control Variables
 - Debt/AV Ratio
 - Median Income
 - Fixed Effects

Variable	Description	Source
spread_bps	Yield spread over synthetic AAA benchmark (Dependent Variable), winsorized at 1st/99th percentile	WRDS Secondary Trades
ever_champ	Binary: 1 if municipality is in the CHAMP cohort (150-point I-Bank resilience project)	NJ CHAMP
is_post_2022	Binary: 1 if trade occurs in 2022 or later (post-CHAMP/STORM Act)	Engineered
slr_exposure_pct	% of municipal assessed value in 4-foot sea-level-rise inundation zone	NJ FloodMapper
time_to_maturity	Years to bond maturity at time of trade	WRDS
debt_to_av	Municipal debt as a share of total assessed property value (fiscal health control)	NJ DCA
median_income	Median household income (in \$10k units in regressions)	US Census ACS
muni_name	Municipality identity (used for fixed effects)	Panel identifier
year	Calendar year of trade (used for fixed effects)	Engineered

Regressions

Difference in Difference

- Panel Data
 - ~187,000 Bonds
- Dependent
 - Yield spread over AAA Proxy
- Treatment
 - July 1, 2022 CHAMP/STORM Act
 - Munis marked with resilient flag
- Controls
 - Muni financial status proxies
 - Fixed Effect Indicators

Baseline Regression (Model A)

$$\text{Spread}_{it} = \alpha + \beta_1 \cdot \text{CHAMP}_i + \beta_2 \cdot \text{Post2022}t + \beta_3 \cdot \text{SLR}_i + \gamma \mathbf{X}_{it} + \varepsilon_{it}$$

where $\mathbf{X}_{it}$ = {time to maturity, debt-to-AV, median income (\$10k)}

Coefficient	Estimate	Interpretation
CHAMP cohort (β_1)	-6.26 bps (p=0.114)	CHAMP towns trade ~6 bps tighter, but not significant — selection bias likely
Post-2022 (β_2)	-29.72 bps (p<0.001)	Market-wide spread compression after 2022
SLR exposure (β_3)	+0.04 bps (p=0.537)	No detectable coastal risk premium in pooled data
Debt-to-AV	-1.32 bps (p=0.055)	Marginally significant — higher debt $\approx$ lower spread (likely endogenous)
Median income	-1.56 bps (p=0.007)	Wealthier towns borrow cheaper

TWFE Diff in Diff (Model B)

$$\text{Spread}_{it} = \alpha_i + \delta_t + \beta \cdot (\text{CHAMP}_i \times \text{Post2022}_t) + \gamma \cdot \text{TTMit} + \varepsilon_{it}$$

where **alpha_i** = municipality fixed effects

Where **deltat** = yearly fixed effects

Coefficient	Estimate	Interpretation
CHAMP × Post-2022 (β)	-32.62 bps (p<0.001)	CHAMP towns saw spreads narrow ~33 bps more than non-CHAMP towns after 2022
Time to maturity (γ)	+4.40 bps (p<0.001)	Each additional year of maturity adds ~4.4 bps (term premium)

Triple Diff in Diff (Model C)

$$\text{Spread}_{it} = \alpha_i + \delta_t + \beta_1(\text{CHAMP}_i \times \text{Post}_t) + \beta_2(\text{Post}_t \times \text{SLR}_i) + \beta_3(\text{CHAMP}_i \times \text{Post}_t \times \text{SLR}_i) + \gamma \cdot \text{TTMit} + \varepsilon_{it}$$

Coefficient	Estimate	Interpretation
CHAMP × Post (β_1)	-35.15 bps (p=0.001)	Baseline CHAMP effect for a town with 0% SLR exposure
Post × SLR (β_2)	-0.09 bps (p=0.490)	No general post-2022 penalty for coastal exposure
CHAMP × Post × SLR (β_3)	+0.19 bps (p=0.497)	Per 1 ppt of SLR exposure — not significant

Questions/Hypotheses

How do markets incorporate towns' sea level rise tax-base exposure (flood risk)?

H_0 : Spreads are uncorrelated with the share of tax base in SRL flooding zones

H_a : Spreads are correlated with the share of tax base in SRL flooding zones

Questions/Hypotheses

Does participation in standardised programs help evaluate bond yields (FEMA CRS)?

H_0 : CRS class has no effect on spreads, holding all other factors constant.

H_a : CRS ratings are systematically associated with spreads, that is resilience metrics account for pricing

Questions/Hypotheses

**Did the state's 2022 funding shift
successfully cause a measurable spread
compression for resilient towns?**

**H_0 : Resilient bonds showed no
significant spread movement after 2022
pivot relative to non-resilient bonds**

**H_a : Resilient bonds experienced stat.
significant spread movement post 2022
due to policy change.**

Results!

Regression Results

Variable	(A) Pooled OLS	(B) TWFE DiD	(C) Coastal DDD
CHAMP Cohort	-6.26	Absorbed by FE	Absorbed by FE
Post-2022	-29.72 ***	Absorbed by FE	Absorbed by FE
SLR Exposure (%)	+0.04	Absorbed by FE	Absorbed by FE
CHAMP × Post-2022	N/A	-32.62 ***	-35.15 ***
Post-2022 × SLR	N/A	N/A	-0.09
CHAMP × Post × SLR	N/A	N/A	+0.19
Debt-to-AV	-1.32 *	Absorbed by FE	Absorbed by FE
Median Income (\$10k)	-1.56 ***	Absorbed by FE	Absorbed by FE
Time to Maturity	+4.42 ***	+4.40 ***	+4.39 ***
R²	0.258	0.384	0.384

Note: *** p<0.01, ** p<0.05, * p<0.1. Fixed Effects (FE) absorb time-invariant variables in Models B and C.

Regression Results

- Flood Risk Sensitivity
 - Debt/Income dominates
 - Heavily confounded by geography/municipal wealth
 - No significant effect of risk on spreads
- Resilience Score Sensitivity
 - Difficult to pinpoint resilience efforts quantitatively
 - Suggest continuing with exact dollar investments toward accepted stormwater mitigation projects
- STORM/CHAMP Effect
 - CHAMP inclusion significantly lowers bond spreads

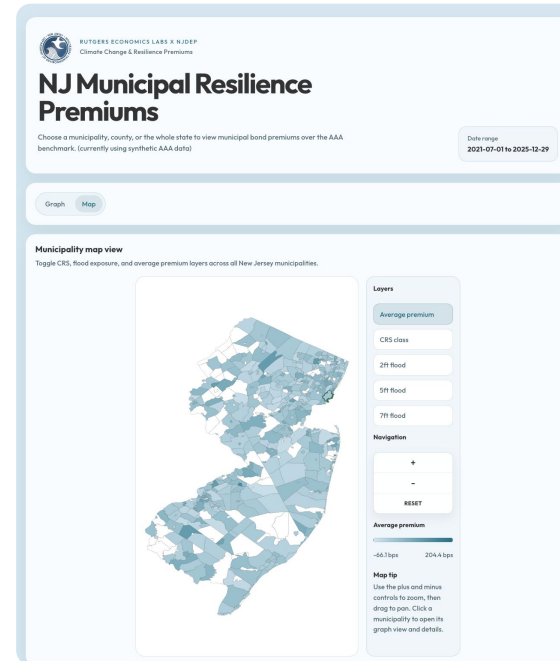
Event Study Results

- Linear pre-trend (-3.73 bps/year; $p = .024$)
- CHAMP towns' spreads converged naturally
 - (+25 bps in 2015 → -23 bps in 2025)

Finding	Evidence	Confidence
CHAMP towns' spreads narrowed ~33 bps more than non-CHAMP after 2022	Model B: $\beta = -32.6***$	Moderate
The CHAMP benefit does not scale with coastal flood exposure	Model C: $\beta_3 = +0.19$, n.s.	No Significance
Pre-treatment convergence is present	Model D: 2015–2016 coefficients significant; linear trend $p=0.024$	Confirmed
Post-2022 treatment effects grow over time	Model D: $-8 \rightarrow -23$ bps (2022–2025)	Strong

Visualization

rutgerseconomics.org/NJDEPClimateRiskSp2026 ← broken right now



Final Report

In Progress

**Project Scope, Significance,
and Forward Outlook**

**Detailed Methodology, Results,
Assumptions, and Limitations**

Thank you,
Questions?

Assumptions

Data Assumptions

- **Tax Base Valuation:** When mapping spatial intersections to calculate "% flooded," we assume the physical footprint of the flood intersection proportionally reflects the actual assessed value of the tax base at risk, ie tax value is evenly distributed over the tax parcel. Our function % flooded $\rightarrow$ % value retained is $V(f) = 1 - (2.5f)^{1/5}$ bounded by 1 and 0.
- **The Policy Pivot Date:** We established a strict structural break on July 1, 2022 (the start of NJ SFY2023). We assume this captures the market reaction to the implementation of the policy, since that markets often price in legislation as early as the drafting phase.
- **Resilience Proxies:** We rely on FEMA CRS scores and NJ CHAMP priority flags as proxies for actual physical resilience. This assumes bureaucratic participation in these programs accurately reflects ground-level infrastructure improvements. We were unable to recover the timeseries of these scores.

Methodology Assumptions

- **The Parallel Trends Assumption:** We assume that in the absence of the 2022 CHAMP/STORM Act, the bond spreads for both resilient (treated) and non-resilient (control) municipalities would have trended together.
- **Information Symmetry:** We assume that municipal bond investors have access to, and actively price in, the spatial flood risk and FEMA CRS class data at the time of issuance. We strongly believe this assumption to be true.
- **Isolating the Climate Premium:** We assume that by controlling for Debt-to-GDP, Median Household Income, and utilizing municipal fixed effects, we have sufficiently isolated the climate risk penalty from standard municipal credit risk.